Beyond Short Names: Emoji Keyword Retrieval across Models and a Bounded Late-Component Test

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Abstract

We use a prospectively specified two-arm design to distinguish cross-model lexical readability from a component-level causal account. Text language models receive emoji as Unicode token sequences rather than rendered pictures. The observational arm crosses 55 emoji and 55 non-emoji Unicode symbols with raw, official-name, word-disjoint CLDR-keyword, and literal codepoint views. Across nine prespecified model–tokenizer configurations, the emoji keyword-minus-codepoint retrieval effect is positive in unrestricted and exact four-view token-count-matched analyses. Every configuration exceeds the registered 0.02 floor, and the four-model partial-conjunction primary passes ($p = 0.0027$). The symbol comparison recovers official names broadly and finds smaller, heterogeneous keyword effects. The passed observational criterion admits a causal arm in Llama 3.2 3B: rank-8 subspaces learned from 128 source emoji are tested at six late attention and MLP outputs on 192 held-out emoji. Engineering controls reproduce the baseline exactly. No primary site satisfies the joint necessity-and-sufficiency criterion under this single-context, low-baseline test. The lexical relation holds in all nine prespecified configurations. The bounded late-component account remains unsupported.

1 Introduction

Readers encounter emoji as compact visual objects. A text language model instead receives encoded Unicode scalars and tokenizer-dependent sequences. An official label may therefore be recoverable because it co-occurs with the character in standards, documentation, or training data. That observation alone cannot establish rendered-glyph perception, a human-like concept, or a localized semantic circuit.

Single-codepoint emoji nevertheless make the recoverability question unusually clean. Under the word-disjointness constraint below, the character shares no substring, no morpheme, and not one ASCII-alphanumeric word with the keyword it must retrieve, so any observed relation must be carried by learned association rather than by surface form. Atomic Unicode characters thereby offer a controlled testbed for how text-only models attach lexical relations to opaque tokens, and a concrete auditing target for multilingual and social-media systems that consume such tokens at scale.

A prospectively specified two-arm design separates two narrower questions. The observational arm asks whether a raw atomic emoji retrieves a word-disjoint CLDR keyword better than its literal `U+...` spelling after exact token-count matching. Conditional on that result, the causal arm asks whether a low-rank late-component intervention identifies a representation that is both necessary and sufficient for the relation. This design yields an auditable distinction between broad lexical

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readability and one simple mechanistic account; it makes no claim of visual perception or human-like emoji understanding.

The study is organized around three safeguards and their corresponding results:

- Four text views and three disjoint item sets separate the raw Unicode surface, official English CLDR name, word-disjoint CLDR keyword, and literal codepoint spelling. The causal source and evaluation sets receive no observational-arm evaluation, although their surfaces may have appeared in earlier studies.
- The observational primary passes: 4/4 core configurations and 9/9 total configurations are positive in unrestricted and exact four-view token-count-matched analyses. The cross-model partial-conjunction value is 0.0027.
- A conditionally admitted causal experiment passes all zero-hook and restoration controls and finds no primary block-24 attention or MLP output that is jointly necessary and sufficient. A secondary block-27 MLP patch raises the composite endpoint through keyword gains and codepoint losses; its necessity criterion fails.

The observational result extends beyond exact-short-name retrieval under the prespecified endpoint. The causal result rejects one prespecified rank-8 late-component mechanism. Distributed, redundant, nonlinear, and item-private explanations remain unresolved.

2 Related work

Textual sources of emoji meaning. Emoji representations have been learned from descriptions [Eisner et al., 2016], linked to lexical sense inventories [Wijeratne et al., 2017], and induced from usage through emoji prediction [Felbo et al., 2017, Barbieri et al., 2018]. These approaches identify descriptions, lexical resources, and context as distinct sources of emoji meaning. We use CLDR names and keywords [Unicode Consortium, 2026] as source-auditable endpoints and ask which item-wise textual relations are recoverable from a raw Unicode input. Co-occurrence with these resources remains a candidate explanation for every positive result.

Probing and tokenizer controls. Probe accuracy can reflect the diagnostic model or its control task [Hewitt and Liang, 2019], and tokenizer choice can materially change model comparisons [Rust et al., 2021]. Our item-wise nearest-neighbor readout avoids a learned diagnostic classifier. Unrestricted and exact four-view token-count-matched tests address length while leaving token identity and tokenizer construction uncontrolled. The readout also differs from whole-representation similarity measures such as CKA [Kornblith et al., 2019]: it asks whether the correct item is the nearest partner across two text views.

Causal intervention. Conventional probing performance need not measure how information is used [Elazar et al., 2021]. Causal mediation, interchange interventions, and model editing instead test whether internal variables affect computation [Vig et al., 2020, Geiger et al., 2021, Meng et al., 2022]. Optimization-based alignment methods go further and learn the intervention subspace itself, either by maximizing interchange accuracy [Geiger et al., 2024, Wu et al., 2023] or through sparse dictionary features [Bricken et al., 2023, Cunningham et al., 2024]. Our component test follows the causal distinction while fixing the subspace by a closed-form cross-covariance rule before any evaluation forward: a prespecified site must pass random-adjusted necessity and sufficiency on a disjoint evaluation set. It therefore rejects one fixed account rather than the best discoverable one, a bound Section 6 records. We retain the resulting negative test without adapting the layer, rank, or endpoint.

3 Prospective design

3.1 Input views and disjoint item sets

For auditability, we retain two internal study labels. **T19-L** denotes the observational lexical-retrieval arm ("L" for lexical), and **T19-C** denotes the conditionally admitted component-causal

Raw scalar†	CLDR `tts`	Keyword	Literal view
U+1F923	rolling on the floor laughing	lmao	`U+1F923`
U+1F928	face with raised eyebrow	skeptic	`U+1F928`
U+00B4	acute accent	tone	`U+00B4`

Table 1: Concrete prespecified views for two emoji and one non-emoji symbol. †The model receives the scalar itself; the table names it by codepoint because the paper typeface need not render every glyph. The literal view is the displayed ASCII string.

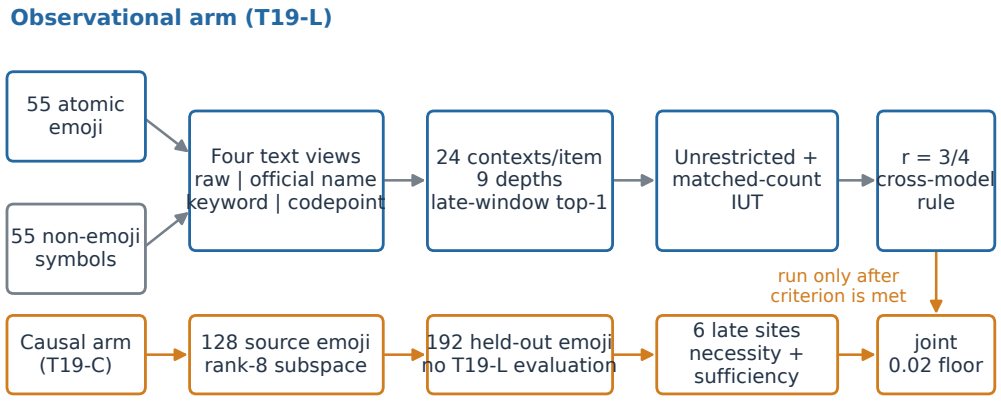


Figure 1: Prospective T19 design. T19-L crosses two glyph kinds with four text views. Only the emoji endpoint enters the registered cross-model primary. T19-C runs only after that criterion is met and uses disjoint source and evaluation emoji. No causal-set item receives a T19-L forward.

arm. They are labels for the two arms defined below; no knowledge of the preceding project sequence is required. Appendix A gives the short decision lineage that led to T19-L without treating earlier exploration as part of its confirmatory evidence.

Throughout, *glyph* means the raw Unicode scalar string supplied to a text model. No raster image is rendered. Atomic emoji are single-codepoint, fully-qualified Emoji 17 entries introduced no later than Emoji 13.1 [Unicode Consortium, 2025]. For each item, the official-name view is the exact English CLDR 48.2 `tts` value; the keyword view is one exact English CLDR `default` keyword. After lowercasing and extracting ASCII `[a-z0-9]+` words, official-name and keyword word sets must be nonempty and disjoint. The fourth view is the uppercase literal codepoint spelling. When several keywords qualify, one deterministic maximum bipartite matching assigns globally unique labels; a fixed SHA-256 edge order breaks ties before any T19 model forward.

The observational set contains 55 emoji and 55 non-emoji Unicode symbols. Symbols are single scalars in a Unicode general category beginning with `S`, have `Emoji=false`, have valid disjoint CLDR keywords, and are absent from every strict-UTF-8, no-NUL repository blob at the prespecified boundary. This is repository-relative freshness, not pretraining novelty. Exact emoji surfaces may have appeared in earlier experiments; the endpoint, keyword assignment, and item-set split are prospective, not the emoji surfaces.

Two further sets contain 128 causal-source and 192 causal-evaluation emoji. All 430 codepoints and 430 keywords are unique across sets. Figure 1 summarizes the separation and the one-way admission rule.

3.2 Observational readout

For each item and view, the prespecified evaluation pipeline appends one of two suffix carriers (`" a"` or `"  |"`) at distances 1, 2, 4, and 8 and records the hidden state at the selected suffix token at nine normalized depths. Batch size is one; no chat template or added special token is used. The primary late window is fixed at depths 0.75, 0.875, and 1.0, yielding 24 contexts per item.

Within each view and context, hidden states are centered across items and each row is L2-normalized. Let $z_i^{v,c}$ denote the resulting vector for item i, view v, and context c. Cross-view top-1 retrieval is

$$R_c(g \to v) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\left[i = \arg\max_j \left\langle z_i^{g,c}, z_j^{v,c} \right\rangle \right]. \tag{1}$$

The per-model primary estimate is

$$\Delta_m = \frac{1}{24} \sum_c \left[R_c(g \to \textsc{keyword}) - R_c(g \to \textsc{codepoint}) \right] \tag{2}$$

on the 55 observational emoji. This is a nearest-neighbor readout from hidden states, not a next-token generation score.

3.3 Token-count intersection and cross-model test

Every model must pass two components. The unrestricted component uses all 55 emoji and permutes keyword correspondence within glyph kind. The matched component retains non-singleton strata sharing the exact four-view token-count signature and permutes keywords only within stratum. Retrieval in that component uses the retained pool ($n = 31$–41 emoji across models), so its chance level is $1/n$; unrestricted and matched magnitudes are not treated as directly commensurable. Official-name and codepoint correspondences stay fixed. Thus the one-sided randomization test asks whether the fixed keyword mapping beats mappings allowed by its null; because the codepoint term is fixed, it is not a paired significance test of keyword–codepoint equality. The registered primary combines that test with the observed comparative requirement $\Delta_m \geq 0.02$ in both components. At $n = 55$, the latter requires at least two net top-1 successes; in the smaller matched pools it requires at least one. Each component uses 19,999 draws, and the model intersection–union test (IUT) p-value is the maximum of the two component values.

The sole cross-model primary uses four prespecified core configurations: Llama 3.2 3B base, Qwen2.5 7B base, OLMo 2 1B base, and Gemma 2 2B IT. It tests the partial conjunction that at least $r = 3$ of $m = 4$ configurations pass [Benjamini and Heller, 2008]. With ordered model IUT values $p_{(1)} \leq \cdots \leq p_{(4)}$, the registered value is $p_{\mathrm{PC}} = \min(1, 2p_{(3)})$. The other five configurations and all symbol or official-name results are secondary and cannot admit the causal arm.

We report percentile intervals from 20,000 item bootstrap replicates as descriptive uncertainty. They are not substituted for the permutation-based primary decision.

3.4 Conditionally admitted causal test

After the observational criterion was committed, T19-C ran in the prespecified Llama 3.2 3B base configuration. The six sites are the attention and MLP component outputs at blocks 21, 24, and 27: the tensors returned by the attention output projection and by the MLP down-projection, hooked before their residual addition. At each site we learned a rank-8 basis from the left singular directions of the centered, row-normalized glyph–keyword cross-covariance on 128 source emoji. The control basis is deterministic rather than ad hoc: sixteen seeded Gaussian directions inside the source span are deflated against the learned basis and orthonormalized, and the first eight are kept. Blocks 24 attention and MLP were primary; blocks 21 and 27 were secondary.

On 192 different evaluation emoji, the two criteria intervene on different forwards. Necessity intervenes on the glyph forward: with P the learned rank-8 projector and μ the source-bank glyph mean at the site, the site output g is replaced by $g - P(g - \mu)$. Sufficiency intervenes on the codepoint forward: the site output c is replaced by $c + P(g - c)$, using the same item's glyph forward, with the replacement cast to BF16 at injection. Analogous operations in the random basis define adjusted effects:

$$N_s = E_s(\text{random removal}) - E_s(\text{learned removal}), \tag{3}$$

$$S_s = E_s(\text{learned patch}) - E_s(\text{random patch}). \tag{4}$$

Both source and evaluation inputs append eight copies of the single-token " |" suffix; the intervention and readout use its first token (distance 1). The endpoint reads the final hidden state of the intervened forward at that token as a query against two fixed 192-row galleries built once from clean, un-intervened forwards: the keyword states and the codepoint states. Query matrix and each gallery are independently centered by their own item means, row-normalized, and scored by cosine top-1 identity, giving E_s = keyword top-1 minus codepoint top-1. The glyph query measures the baseline and necessity conditions. The patched codepoint query measures sufficiency; there the codepoint term is self-retrieval and equals 1 at baseline, so a patch can raise E_s either by real keyword gains or by destroying self-retrieval, and Section 4 reports the two components separately. E_s shares the item-wise contrast definition with T19-L, while using a different item pool and one fixed context; it is distinct from the observational 24-context average and its token-count-matched component. A site must have both adjusted effects and both lower 95% item-bootstrap bounds above 0.02, pass both one-sided sign-flip tests (99,999 draws), pass zero-hook and remove-restore engineering controls, and survive Holm correction across the two primary sites [Holm, 1979]. No site, rank, depth, carrier, subset, or endpoint may be replaced after seeing the result.

3.5 Freeze, admission, and audit chain

The prospective claims above are enforced mechanically rather than asserted. A freeze step hashes every protocol asset (configuration, item selections, acquisition and analysis code, preflight receipts) into a manifest that can only be built while two zero-forward preflight receipts exist, and an audit command re-hashes every asset and fails unless the freeze commit is pushed to the repository default branch. The causal arm opens through a separate admission manifest that binds the freeze-manifest hash, the observational analysis-receipt hash, both causal item-bank hashes, and forward counters required to read zero at admission time. The causal executor rehashes its own file at every invocation and refuses to run if it differs from the hash bound at admission. The evidence archive contains the freeze and admission manifests and the independent validation receipts, so this ordering is checkable from the submitted package rather than testimonial.

4 Results

4.1 A lexical relation beyond the official short name

The registered cross-model primary passed. All four core configurations were positive; the third-smallest core IUT value was 0.00135, yielding $p_{\mathrm{PC}} = 2(0.00135) = 0.0027$. All nine prespecified configurations were positive and exceeded the registered 0.02 effect-size floor in both unrestricted and matched analyses (Figure 2; exact values in Appendix C). Model IUT values ranged from 0.00005 to 0.00185. The smallest unrestricted effect, Qwen2.5 0.5B at $+0.03636$, is equivalent to an average margin of two net top-1 matches per 55-item context. A separate validator that does not import the acquisition analyzer reloaded the compact predictions, reconstructed all 36 glyph-kind by component randomization tests, and reproduced the primary decision.

Across models, mean emoji keyword effects were $+0.0979$ unrestricted and $+0.1084$ matched. Mean official-name effects were larger, $+0.1440$ and $+0.1718$. Thus, retrieval extends beyond the exact official name, although the official name remains the stronger key on average. The alternative label is an exact CLDR keyword, not a human-rated synonym or paraphrase; training-data co-occurrence could still fully explain the relation. Descriptively, the unrestricted effect also rises monotonically with parameter count inside the one family sampled at four sizes (Qwen2.5: $+0.036$, $+0.053$, $+0.071$, $+0.182$ from 0.5B to 7B). The roster remains confounded across families, so this ordering is description, not a registered scale result.

Depth was secondary. The nine recorded fractions were $0, 0.125, \ldots, 1.0$; only 0.75, 0.875, and 1.0 formed the primary window. Across models and contexts, the unrestricted contrast was $+0.0409$ at depth 0, then $+0.0864$, $+0.1152$, and $+0.0922$ at the three primary depths, whose mean is the $+0.0979$ reported above. Individual configurations had different onset and peak depths, so these data do not support one universal trajectory.

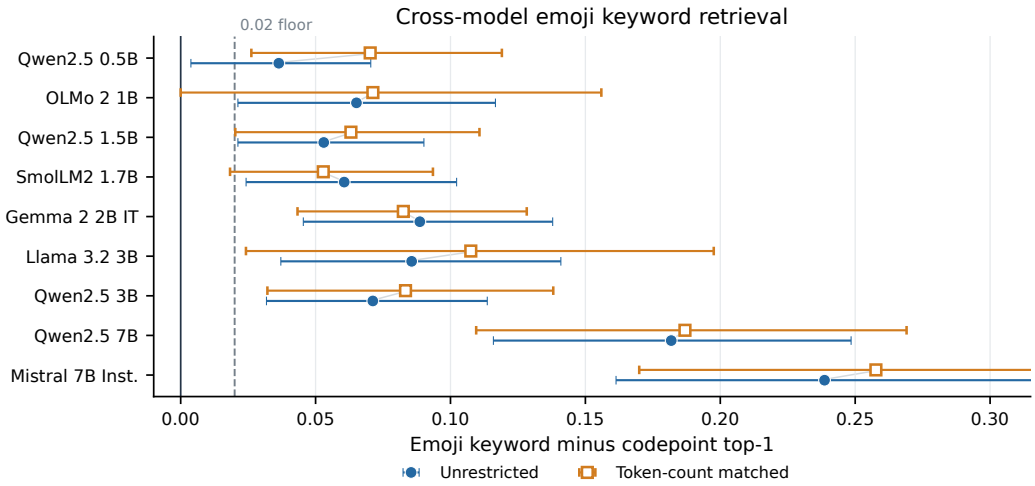


Figure 2: Emoji keyword-minus-codepoint top-1 effects in the registered roster order. Bars are descriptive 95% item-bootstrap intervals (20,000 replicates); randomization tests determine the primary decision. The 0.02 floor applies to point estimates. Unrestricted (U) and matched (M) analyses use different retained pools, so their magnitudes are not directly commensurable. Every displayed point estimate exceeds the floor.

4.2 A named-symbol control limits the emoji-specific claim

Non-emoji symbols showed mean keyword effects of $+0.0236$ unrestricted and $+0.0153$ matched, compared with $+0.0979$ and $+0.1084$ for emoji. The emoji-minus-symbol point estimate was positive in all nine configurations in both components. Symbol keywords were heterogeneous: Qwen2.5 3B and 7B were negative in both components. Official-name retrieval exceeded keyword retrieval for symbols in 9/9 configurations in both components, with mean symbol official-name effects of $+0.0750$ and $+0.0821$ (Figure 3).

This comparison cannot establish an emoji-specific faculty. The panels are not matched on training frequency, lexical specificity, Unicode category, visual familiarity, or token identity. Its role is to prevent the incorrect inference that only emoji acquire readable names.

4.3 No prespecified late-component site passes both causal criteria

Before intervention, the final-hidden glyph endpoint on the 192-item causal evaluation set was $E_s = 0.07292$ (keyword top-1 0.07292, i.e. 14 of 192 items; codepoint top-1 0); the zero-hook engineering check reproduced this baseline identically at all six sites. The 0.02 decision floor was therefore 27% of this baseline contrast, and the design contained no positive intervention control that calibrated attainable adjusted necessity at the primary sites. All six causal sites passed their engineering checks: zero-hook and remove-restore conditions reproduced the baseline final hidden state with exactly zero RMS difference. Bitwise equality is the expected pass state, not a numerical coincidence: baseline forwards reuse the exact intervention batch geometry, so any nonzero residue would indicate hook or batching infidelity. Nevertheless, neither primary block-24 site passed. Block-24 attention had adjusted necessity 0.0000 and sufficiency -0.00521; block-24 MLP had $+0.00521$ and 0.0000. Their site IUT values were 1.0 and 0.74912, respectively. None of the six sites passed both components (Figure 4; exact values in Appendix D).

The secondary block-27 MLP patch produced adjusted sufficiency $+0.07812$ (95% interval $[+0.03646, +0.11979]$; one-sided $p = 0.00009$, uncorrected, with twelve such component tests run in total). Necessity was -0.01562 (interval $[-0.04167, +0.00521]$, $p = 0.96895$), giving a site IUT of 0.96895. The learned patch raised keyword top-1 from $1/192$ to $4/192$ while lowering codepoint self-retrieval from $192/192$ to $180/192$. The keyword gain accounts for 3 of the composite endpoint's 15 additional correct contrasts; the remaining 12 arise from lost codepoint self-retrieval.

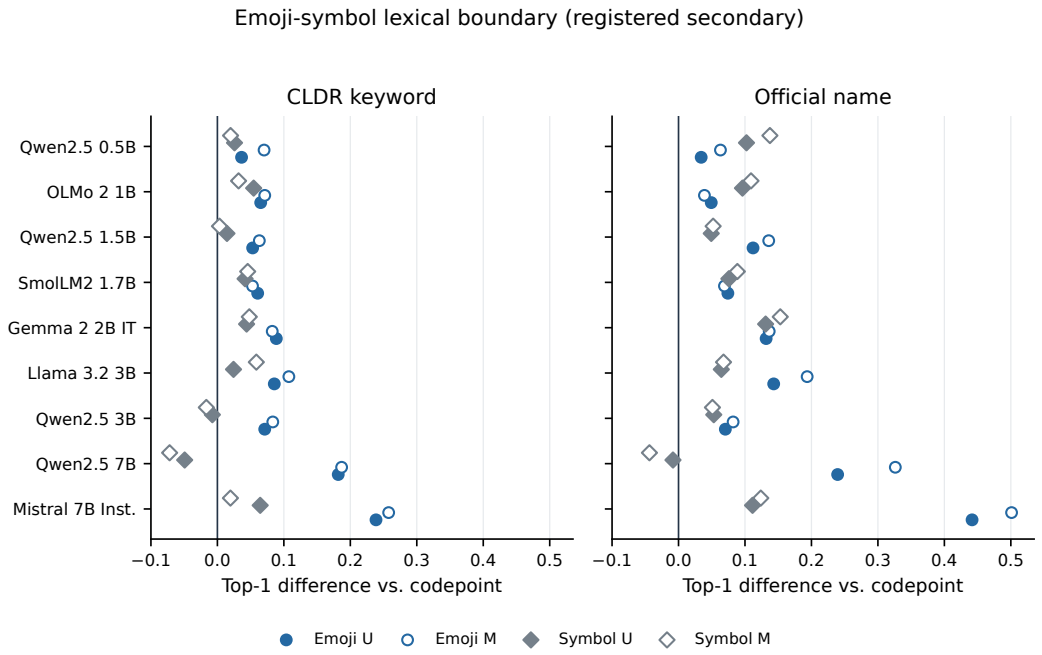


Figure 3: Registered secondary comparison. Each row shows unrestricted (U) and token-count-matched (M) effects for the same prespecified model. Official-name retrieval is not exclusive to emoji; disjoint-keyword effects are stronger and more consistent for the prespecified emoji panel. Glyph category, frequency, label quality, and token identity are not exchangeable, so this is a boundary result rather than an emoji-only semantic claim.

This pattern is consistent with broad representational disruption and does not identify a necessary keyword mechanism.

5 Discussion

The two arms answer different questions. Under a fixed text-only readout, T19-L recovers the same item across raw emoji and a word-disjoint CLDR keyword more reliably than across raw emoji and a literal codepoint string. Exact token-count matching makes a pure length account insufficient. The symbol panel places the official-name effect within a broader Unicode phenomenon and shows graded, rather than categorically emoji-only, keyword breadth.

T19-C evaluates a narrower mechanistic hypothesis: one shared rank-8 late attention or MLP component learned from a source set and tested on a separate evaluation set. None of the prespecified sites passes joint necessity and sufficiency. Aggregate linear readability therefore remains compatible with the absence of a single shared rank-8 late component. Distributed, redundant, nonlinear, and item-private accounts remain open.

The block-27 MLP asymmetry sharpens that distinction. Its patch moved the composite endpoint mainly by degrading codepoint self-retrieval. Requiring necessity, sufficiency, random-subspace adjustment, restoration, and a disjoint evaluation set prevents that isolated movement from being upgraded to a circuit claim.

6 Limitations and broader impact

Measurement. CLDR keywords are standardized annotations, not human judgments of synonymy, natural naming, specificity, or familiarity. Retrieval is nearest-neighbor geometry at selected suffix tokens, not generation or task behavior. Inputs are Unicode text; no rendered glyph is perceived. Exact token-count matching does not match token identities or equalize tokenizer construction.

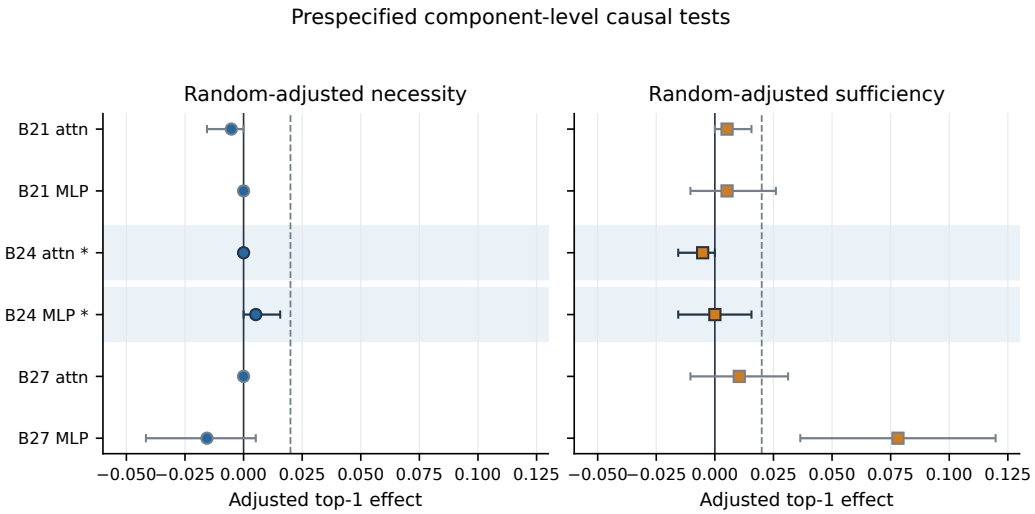


Figure 4: Random-adjusted causal effects with descriptive 95% item-bootstrap intervals over 192 emoji held out from T19-L. Shaded rows and asterisks mark the two primary block-24 sites. Solid vertical lines show zero; dashed lines show the registered 0.02 effect-size floor. No site has both necessity and sufficiency above the floor with positive lower bounds.

Generalization. The nine model–tokenizer configurations are a prespecified, convenience-constrained roster spanning several families and sizes [Yang et al., 2024, Groeneveld et al., 2024, Dubey et al., 2024, Gemma Team et al., 2024, Jiang et al., 2023, Ben Allal et al., 2025]. Model family, parameter count, training data, tokenizer, and instruction tuning are confounded, so the study supports neither a scale law nor a model-population estimate. Exact emoji surfaces are not pretraining-novel. The symbol panel is repository-text-fresh but not a pretraining holdout and is not exchangeable with the emoji panel.

Mechanism. The negative causal result covers one Llama 3.2 3B checkpoint, rank 8, one carrier and distance, two primary and four secondary sites, and one final-hidden endpoint. That endpoint has less aggregation and lower baseline signal than the observational primary. Its exact engineering checks verify hook fidelity, while the absence of a prespecified positive intervention control leaves power and attainable dynamic range uncalibrated. The result does not exclude other causal representations: optimized interchange subspaces [Geiger et al., 2024, Wu et al., 2023], sparse feature bases [Bricken et al., 2023, Cunningham et al., 2024], distributed multi-site carriers, and nonlinear readouts were not evaluated. A future study should preregister such a positive control together with those alternatives on a new source/evaluation split rather than selecting a method on the present evaluation set.

Reproducibility and impact. The anonymous compact archive binds configurations, selections, summaries, tests, validators, license notices, and the rendered paper. It omits model-dependent raw arrays retained in execution scratch. Exact acquisition duration and peak memory were not receipt-bound and are not reconstructed after the fact. Better Unicode auditing could reveal hidden representation failures in multilingual or social-media systems. Conversely, treating CLDR correspondence as a person's meaning, sentiment, culture, or identity could erase contextual variation and enable unjustified profiling. The measurements should not be used for such inference. No personal data, human subjects, or user messages are used.

7 Conclusion

Across nine prespecified model–tokenizer configurations, raw atomic emoji retrieve word-disjoint CLDR keywords better than literal codepoints after exact four-view token-count matching. A named-symbol comparison bounds the specificity of that observation. A separately partitioned causal ex-

periment finds no primary rank-8 late component that is jointly necessary and sufficient. Together, these results establish broad lexical readability under a text-only metric and reject the prespecified sparse late-component account. Visual understanding and a general emoji circuit remain outside the evidence.

References

Francesco Barbieri, Luis Espinosa-Anke, Jose Camacho-Collados, Steven Schockaert, and Horacio Saggion. Interpretable emoji prediction via label-wise attention LSTMs. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 4766–4771. Association for Computational Linguistics, 2018. doi: 10.18653/v1/D18-1508. URL https://aclanthology.org/D18-1508/.

Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, et al. SmolLM2: When smol goes big — data-centric training of a small language model. *arXiv preprint arXiv:2502.02737*, 2025. URL https://arxiv.org/abs/2502.02737.

Yoav Benjamini and Ruth Heller. Screening for partial conjunction hypotheses. *Biometrics*, 64(4): 1215–1222, 2008. doi: 10.1111/j.1541-0420.2007.00984.x.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nicholas L. Turner, Cem Anil, Carson Denison, Amanda Askell, et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features/index.html.

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2309.08600.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL https://arxiv.org/abs/2407.21783.

Ben Eisner, Tim Rocktäschel, Isabelle Augenstein, Matko Bošnjak, and Sebastian Riedel. emoji2vec: Learning emoji representations from their description. In *Proceedings of the Fourth International Workshop on Natural Language Processing for Social Media*, pages 48–54. Association for Computational Linguistics, 2016. doi: 10.18653/v1/W16-6208. URL https://aclanthology.org/W16-6208/.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359. URL https://aclanthology.org/2021.tacl-1.10/.

Bjarke Felbo, Alan Mislove, Anders Søgaard, Iyad Rahwan, and Sune Lehmann. Using millions of emoji occurrences to learn any-domain representations for detecting sentiment, emotion and sarcasm. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 1615–1625. Association for Computational Linguistics, 2017. doi: 10.18653/v1/D17-1169. URL https://aclanthology.org/D17-1169/.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 9574–9586, 2021. URL https://proceedings.neurips.cc/paper/2021/hash/4f5c422f4d49a5a807eda27434231040-Abstract.html.

Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah D. Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Proceedings of the Third Conference on Causal Learning and Reasoning*, volume 236 of *Proceedings of Machine Learning Research*. PMLR, 2024. URL https://proceedings.mlr.press/v236/geiger24a.html.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024. URL https://arxiv.org/abs/2408.00118.

Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Harsh Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, et al. OLMo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*, 2024. URL https://arxiv.org/abs/2402.00838.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 2733–2743. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1275. URL https://aclanthology.org/D19-1275/.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023. URL https://arxiv.org/abs/2310.06825.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR, 2019. URL https://proceedings.mlr.press/v97/kornblith19a.html.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372, 2022. URL https://papers.nips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.

Phillip Rust, Jonas Pfeiffer, Ivan Vulić, Sebastian Ruder, and Iryna Gurevych. How good is your tokenizer? on the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*, pages 3118–3135. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.acl-long.243. URL https://aclanthology.org/2021.acl-long.243/.

Unicode Consortium. Unicode emoji, version 17.0. Unicode Standard Annex data file, 2025. URL https://www.unicode.org/Public/17.0.0/emoji/emoji-test.txt.

Unicode Consortium. Unicode common locale data repository, version 48.2, 2026. URL https://cldr.unicode.org/.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://papers.neurips.cc/paper_files/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html.

Sanjaya Wijeratne, Lakshika Balasuriya, Amit Sheth, and Derek Doran. EmojiNet: An open service and api for emoji sense discovery. In *Proceedings of the Eleventh International AAAI Conference on Web and Social Media*, pages 437–446. AAAI Press, 2017. doi: 10.1609/icwsm.v11i1.14857. URL https://ojs.aaai.org/index.php/ICWSM/article/view/14857.

Zhengxuan Wu, Atticus Geiger, Thomas Icard, Christopher Potts, and Noah D. Goodman. Interpretability at scale: Identifying causal mechanisms in alpaca. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2305.08809.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. URL https://arxiv.org/abs/2412.15115.

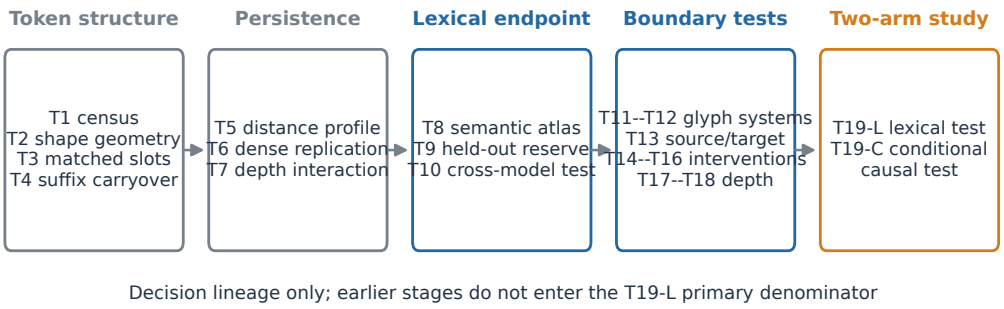


Figure 5: Exploration-to-confirmation lineage. Early studies exposed token-structure and persistence explanations; later studies narrowed the endpoint and causal test. Arrows record design decisions, not evidence pooling. Only the T19-L primary supports the confirmatory lexical claim in this paper.

A Exploration path and decision lineage

T19-L is the nineteenth internal study label in a longer audit trail, not a concept that readers are expected to know in advance. The concise lineage below includes only results that changed the next design decision. These studies motivated T19-L. They are not pooled with its preregistered primary, and their exploratory or failed-confirmation status is unchanged.

Figure 5 groups the sequence by the question each stage resolved. The labels trace design decisions rather than a cumulative evidence pipeline.

Initial screen and matched controls (baseline and Milestone 2). A colored-shape steering screen produced 25 positive results among 36 layer–seed–dose configurations. An exact tokenization-matched follow-up supported one tested layer contrast and failed to support the other. We made tokenization part of the estimand, retained negative configurations, and stopped treating a pooled steering score as one stable semantic effect.

Token structure and short-range carryover (T1–T4). T1, the *multifamily tokenizer census*, found no universal token-count and prefix match for the fixed colored-shape panel. T2, *multifamily colored-shape geometry*, yielded no dual-model recurrence at the selected midpoint. T3, *five-family matched-slot geometry*, found recurrence only in OLMo, where the exact shared slot token was already a sufficient explanation. T4, *common-suffix carryover*, showed that the OLMo pattern persisted one common token later. The sequence converted tokenizer structure from a nuisance variable into an explicit competing explanation.

Persistence, depth, and the lexical turn (T5–T8). T5, the *carrier-distance profile*, and T6, the *dense CUDA carrier-distance replication*, both failed their prespecified distance-attenuation conjunctions. T7, *Mistral base depth-by-distance*, passed an early-versus-late interaction in a model and endpoint selected after T6, so it remained prospective within that configuration rather than independent confirmation. T8, the *Mistral semantic atlas*, then mapped 842 discovery emoji and found a large descriptive name-over-codepoint contrast. That atlas supplied the endpoint later tested on the untouched T9 reserve.

Lexical exploration and failed reserves (T9–T12). A held-out Mistral configuration first supported name-over-codepoint retrieval. A nine-configuration follow-up retained statistical signal and failed its registered cross-model effect-size rule. Broader glyph-system exploration then suggested Unicode-composition hypotheses; neither survived testing on separately reserved items. We kept both failures, dropped the generic composition claim, and returned to a narrower lexical question.

Specificity decomposition and first causal localization (T13–T14). T13 separated structured source directions from generic target susceptibility on already exposed panels and remained exploratory. T14 then paired a signed six-model intervention grid with a separately specified selected-site test. The broad response pattern was heterogeneous across model, depth, source form, null family, and sign; the selected Llama component failed the joint causal criterion on a new evaluation set. This combination motivated the response-shape tests and stricter causal separation in T15–T16.

Readable responses and unsuccessful causal tests (T15–T16). An emoji–word response pattern replicated prospectively, and prespecified forced-choice associations were detectable. Yet a subspace intervention and a multi-layer transplantation each failed their stronger causal criteria. This separated readout from implementation and motivated the present joint necessity–sufficiency criterion.

Depth mapping and fresh confirmation (T17–T18). Across 292 previously exposed atomic emoji, name-minus-codepoint retrieval was positive in all nine model configurations at middle-to-late depths. Peaks were model-specific, and final item retrieval (9/9) did not require consistent whole-representation CKA alignment (4/9 configurations). We therefore prespecified a common late window, replaced the population, and made no shared-transition or universal-manifold claim. On 192 repository-fresh composite emoji, the official-name contrast then passed in 4/4 core configurations and 8/9 configurations overall after exact token-count matching. This justified asking whether the relation extends beyond the official short name, while adding a named non-emoji control and reserving new item sets for causality. This T18 composite set is distinct from the 192 atomic emoji in the T19-C evaluation set.

T19-L and T19-C (current confirmatory and conditional arms). T19-L replaces the sole label endpoint with a word-disjoint CLDR keyword and adds literal-codepoint and named-symbol boundaries. T19-C tests the prespecified rank-8 component account on disjoint source and evaluation sets. The present paper reports the resulting pair of differently gated results: broad lexical retrieval passes, while the simple late-component mediator does not.

This lineage explains why the final design contains four text views, exact token-count strata, a non-emoji symbol panel, three disjoint item sets, and a necessity–sufficiency intersection. It also explains what is absent: no favorable prior layer, glyph system, or causal site was promoted into the T19-L primary. Only that prespecified primary supports the paper's confirmatory lexical claim.

B Prespecified model roster

Table 2 reports the exact model artifacts and revisions used by the prespecified observational roster. The Llama role uses the pinned BF16 mirror artifact specified by the protocol. Despite its `mlx-community` identifier, the recorded run loaded its safetensors through the Transformers/CUDA backend; we make no upstream bit-identity claim. Model cards and upstream license terms are part of the submission-archive license audit. The cards declare Apache-2.0 for the Qwen2.5 0.5B, 1.5B, and 7B configurations, OLMo, SmolLM2, and Mistral; Qwen2.5 3B uses the non-commercial Qwen Research License, Gemma uses the Gemma Terms of Use, and the Llama mirror uses the Llama 3.2 Community License. No checkpoint or tokenizer file is redistributed.

Role	Artifact identifier	Revision
Qwen2.5 0.5B base	`Qwen/Qwen2.5-0.5B`	`060db6499f32`
OLMo 2 1B base	`allenai/OLMo-2-0425-1B`	`a1847dff3500`
Qwen2.5 1.5B base	`Qwen/Qwen2.5-1.5B`	`8faed761d45a`
SmolLM2 1.7B base	`HuggingFaceTB/SmolLM2-1.7B`	`effd688a1292`
Gemma 2 2B IT	`google/gemma-2-2b-it`	`299a8560bedf`
Llama 3.2 3B base	`mlx-community/Llama-3.2-3B-bf16`	`60a99aaf4316`
Qwen2.5 3B base	`Qwen/Qwen2.5-3B`	`3aab1f1954e9`
Qwen2.5 7B base	`Qwen/Qwen2.5-7B`	`d14972939875`
Mistral 7B Inst. v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	`c170c708c41d`

Table 2: Pinned artifact identifiers and the first 12 hexadecimal characters of their complete pinned revisions. Full revisions are stored in the submission manifest.

C Exact observational results

Model configuration	U effect	M effect	Model IUT p	Pass
Qwen2.5 0.5B	0.0364	0.0702	0.00055	yes
OLMo 2 1B	0.0652	0.0712	0.00135	yes
Qwen2.5 1.5B	0.0530	0.0631	0.00185	yes
SmolLM2 1.7B	0.0606	0.0528	0.00005	yes
Gemma 2 2B IT	0.0886	0.0825	0.00005	yes
Llama 3.2 3B	0.0856	0.1075	0.00135	yes
Qwen2.5 3B	0.0712	0.0833	0.00005	yes
Qwen2.5 7B	0.1818	0.1869	0.00005	yes
Mistral 7B Inst.	0.2386	0.2577	0.00005	yes

Table 3: Exact reporting values for the registered emoji keyword-minus-codepoint endpoint. U is unrestricted; M is exact four-view token-count matched. The model IUT is the maximum component randomization p-value.

Matched emoji counts ranged from 31 to 41. The matched component retained at least 63 observational items total in every configuration, with at least 31 emoji and 32 symbols; the unrestricted component used all 55 emoji and 55 symbols. The acquisition comprised $9 \times 110 \times 4 \times 2 = 7{,}920$ batch-one forwards (configurations $\times$ items $\times$ views $\times$ carriers). Four distances and nine depths produced $7{,}920 \times 4 \times 9 = 285{,}120$ readout vectors in 45 sealed shards.

D Exact causal results

Block	Site	Primary	N_s	p_N	S_s	p_S	Site IUT
21	attention	no	-0.00521	1.00000	0.00521	0.49983	1.00000
21	MLP	no	0.00000	1.00000	0.00521	0.49760	1.00000
24	attention	yes	0.00000	1.00000	-0.00521	1.00000	1.00000
24	MLP	yes	0.00521	0.49989	0.00000	0.74912	0.74912
27	attention	no	0.00000	1.00000	0.01042	0.31511	1.00000
27	MLP	no	-0.01562	0.96895	0.07812	0.00009	0.96895

Table 4: Random-adjusted necessity (N_s) and sufficiency (S_s). All six engineering checks passed. Holm correction applies across the two primary sites and did not change either decision; no site passed the component-level intersection.

The causal run used $128 \times 2 = 256$ source forwards. For each of 192 evaluation items, three baseline-view forwards plus two batched intervention forwards at each of six sites yielded $192 \times 15 = 2{,}880$ evaluation forwards; seven intervention rows per site yielded $192 \times 6 \times 7 = 8{,}064$ rows. Each of 12 component tests used 99,999 sign flips and 20,000 item-bootstrap replicates. An independent validator rehashed source and evaluation arrays, reconstructed all 12 tests and intervals, verified learned and random subspace relations, and reproduced six engineering passes and zero positive sites.

E Execution, licensing, and artifact boundary

Both observational and causal acquisition used a fixed Transformers/CUDA BF16 environment and an NVIDIA GeForce RTX 5090. Model artifacts, tokenizer replay receipts, selection files, analysis summaries, randomization seeds, bootstrap seeds, and validators are checksum-bound. Figure inputs in the paper are only the compact validated model-configuration and causal-site JSON summaries; a generated manifest hashes both inputs and every plotted PDF/PNG output.

The anonymous evidence archive is a deterministic whitelist-only zip. Its manifest records every payload path, byte count, and SHA-256; a validator rejects unsafe paths, author or local-path tokens, model-weight formats, and raw array formats. The archive includes the paper, compact results, selected item sets, code, validation receipts, and paired license/compute/ethics reports. It omits model-dependent activation arrays, learned causal subspaces, the full intervention ledger, model weights,

and tokenizer files. Those raw scientific objects remain bound by run receipts in execution scratch. The package thus supports compact-summary and figure audit, not self-contained raw replay.

The observational acquisition used 7,920 batch-one forwards and produced 285,120 readout vectors. Causal acquisition used 256 source forwards, 2,880 evaluation forwards, and 8,064 intervention rows. No training or generation was performed. Exact wall-clock time, peak memory, energy use, and total exploratory compute were not preserved in the T19 compact receipts; the compute checklist therefore remains negative rather than substituting older run timings.

Selected Unicode 17 and CLDR 48.2 rows are used under Unicode License v3 and travel with its copyright and permission notice. New project code is Apache-2.0. Restricted checkpoint assets are referenced by pinned revision and upstream terms rather than redistributed. In particular, the Qwen2.5 3B configuration is used only for non-commercial research under the Qwen Research License.

We reviewed the study against the NeurIPS Code of Ethics. It uses no human participants, personal data, private messages, or scraped user content. LLM coding and research assistants supported protocol drafting, code generation, manuscript editing, and one author-side fresh-eyes review; they did not provide labels, choose outcomes after T19 execution, or serve as the scientific evaluator. The authors retained execution authority and approved the prespecified endpoints and final claims; numerical evidence and validators were checksum-bound.

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5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The study depends on code, so N/A is inapplicable. A checksum-bound anonymous compact artifact is supplied, but no public repository is available at submission and the package omits model-dependent raw arrays required for exact replay. Full code and eligible data are planned for public release after review.

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Answer: [Yes]

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